

Normalized Cuts and Image Segmentation

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What are we solving?

- Image segmentation = dividing an image into meaningful regions or objects
- Traditional local methods only consider nearby pixel information
- Such methods often fail to capture the global structure of the image
- Goal of segmentation:
 - High similarity within groups
 - Low similarity between different groups

Approach: Represent the image as a weighted graph and partition it optimally.

Graph Construction

- Each pixel is represented as a node in the graph
- Edges connect nearby pixels
- Edge weights measure similarity between pixels
- Similarity depends on:
 - Intensity or color similarity
 - Spatial distance between pixels

$$w(i, j) = \exp \left(-\frac{\|F(i) - F(j)\|^2}{\sigma_I^2} \right) \cdot \exp \left(-\frac{\|X(i) - X(j)\|^2}{\sigma_X^2} \right)$$

Interpretation: Nearby pixels with similar appearance receive stronger connections.

Minimum Cut Problem

$$\text{cut}(A, B) = \sum_{i \in A, j \in B} w(i, j)$$

- Minimum cut tries to minimize the connection between two partitions
- Computationally efficient
- However, it often separates very small isolated regions
- This produces unbalanced segmentation results

Problem: Standard min-cut is biased toward small partitions.

Need for Normalized Cut

$$Ncut(A, B) = \frac{cut(A, B)}{assoc(A, V)} + \frac{cut(A, B)}{assoc(B, V)}$$

- Normalized Cut considers both separation and total association
- Penalizes very small isolated groups
- Produces more balanced partitions
- Measures relative disconnection instead of absolute disconnection

Key Idea: Good segmentation should minimize inter-group similarity while maximizing intra-group similarity.

Mathematical Formulation

- Represent graph using matrices:

W = weight matrix, D = degree matrix

$$D_{ii} = \sum_j w(i, j)$$

- Define partition using vector x

$$\min \frac{y^T (D - W) y}{y^T D y}$$

- Segmentation is formulated as an optimization problem
- The expression is known as the Rayleigh Quotient
- Exact minimization is computationally NP-hard
- Therefore, the problem is relaxed into a continuous form

Goal: Find the partition that minimizes the normalized cut value.

$$(D - W)y = \lambda Dy$$

- The optimization problem becomes a generalized eigenvalue problem
- The smallest eigenvalue corresponds to a trivial solution
- The second smallest eigenvector provides the optimal partition
- This forms the basis of spectral clustering

Insight: Image segmentation is transformed into a linear algebra problem.

General Algorithm

- ① Construct weighted graph from image
- ② Compute similarity and degree matrices
- ③ Solve generalized eigenvalue problem
- ④ Use eigenvectors to partition the graph
- ⑤ Recursively subdivide regions if required

Outcome: Pixels with similar properties are grouped into the same segment.

Two-Way Recursive Cut

- Uses the second smallest eigenvector for partitioning
- Splits the graph into two regions at each step
- Threshold selected to minimize Ncut value
- Process is recursively repeated on sub-regions

Advantage:

- Simple implementation
- Effective for hierarchical segmentation

K-Way Partitioning using Clustering

- Instead of recursively applying two-way cuts, multiple eigenvectors are used together
- The top eigenvectors act as feature vectors for each pixel
- These eigenvectors form a higher-dimensional representation of the image
- A clustering algorithm such as k-means is then applied
- This produces an initial oversegmentation into multiple groups

Key Idea:

- Multiple eigenvectors capture richer structural information
- Segmentation is obtained simultaneously instead of repeated binary splitting

Condensed Graph and Recursive Refinement

- After initial K-way segmentation, each segment becomes a node in a condensed graph
- Edge weights between segments represent inter-group similarity
- The condensed graph is much smaller and easier to process
- Recursive Two-Way Normalized Cut is then applied again on this condensed graph
- This further refines and improves the segmentation quality

Advantages:

- Reduces computational complexity
- Preserves global image structure
- Produces more balanced and meaningful partitions

Experiment Setup

- Objective: Compare Recursive Two-Way Cut and K-Way Cut
- Input image: Lion image
- The image is converted into a weighted graph
- Pixel similarity is computed using intensity and spatial distance
- Experiments demonstrate how segmentation differs between both methods



Figure: Figure of Lion that is used in the experiment

Processing Steps

- Construct similarity matrix W
- Compute degree matrix D
- Solve generalized eigenvalue equation
- Extract important eigenvectors
- Apply:
 - Recursive Two-Way Cut
 - Simultaneous K-Way Cut
- Compare resulting image segments

Comparison: Two-Way vs K-Way

- **Recursive Two-Way Cut**

- Splits image into two regions repeatedly
- Simpler and computationally efficient
- May lose global structural information

- **K-Way Cut**

- Uses multiple eigenvectors simultaneously
- Produces richer and more balanced segmentation
- Better captures complex image structures

Observation: K-Way segmentation produced more informative object separation in the lion image experiment.

Two Way recursive cut Vs K-way Cut

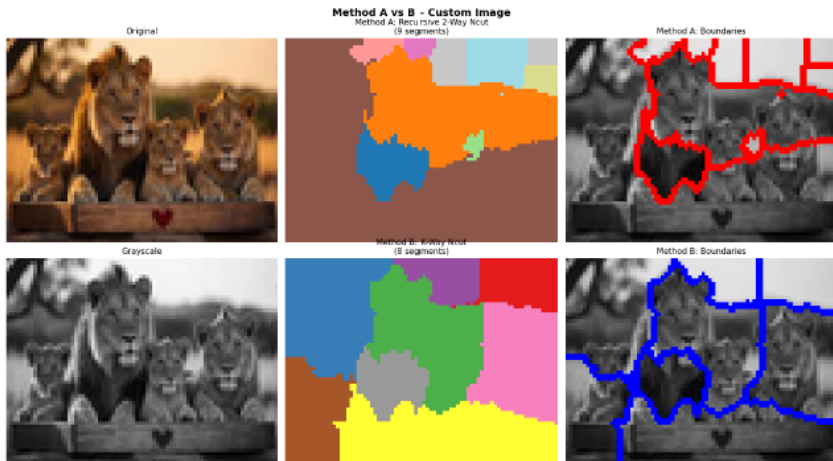


Figure: Demonstrating the difference between Two Way recursive cut Vs K-way Cut

Main Conclusions

- Normalized Cut provides a global criterion for image segmentation
- Avoids the small partition problem of minimum cut
- Converts segmentation into a spectral graph problem
- Eigenvectors help identify meaningful image regions
- Experimental results show effective segmentation using both Two-Way and K-Way methods

Key Takeaway: Graph theory and spectral methods provide powerful tools for image segmentation.